

Quiz 4 — Short Solutions

Exercise 1. $A = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 4 \\ -1 & 0 \end{pmatrix}$.

Solution. (a) $A + B = \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix}$, $AB = \begin{pmatrix} 4 & 4 \\ -3 & 0 \end{pmatrix}$.

$$A + B = \begin{pmatrix} 3 & 2 \\ -1 & 3 \end{pmatrix}, \quad AB = \begin{pmatrix} 4 & 4 \\ -3 & 0 \end{pmatrix}$$

(b) $(AB)^\top = \begin{pmatrix} 4 & -3 \\ 4 & 0 \end{pmatrix} \neq AB \Rightarrow \boxed{AB \text{ is not symmetric}}.$

(c) $\det(AB) = 4 \cdot 0 - 4(-3) = 12 \neq 0 \Rightarrow (AB)^{-1} \text{ exists:}$

$$(AB)^{-1} = \frac{1}{12} \begin{pmatrix} 0 & -4 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{3} \\ \frac{1}{4} & \frac{1}{3} \end{pmatrix}.$$

Exercise 2.
$$\begin{cases} x + 2y + 3z = 1 \\ 2x - y + z = 2 \\ x - 2y + 2z = -1 \end{cases}$$

Solution. (a) $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 1 & -2 & 2 \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\vec{b} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$; $\boxed{A\vec{x} = \vec{b}}.$

(b) $\det A = -15 \neq 0 \Rightarrow \text{rank } A = 3 \text{ and the system has a } \boxed{\text{unique solution}}.$

(c) Cramer for y : replace the 2nd column of A by $\vec{b}$:

$$A_y = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 2 & 1 \\ 1 & -1 & 2 \end{pmatrix}, \quad \det A_y = -10, \quad \boxed{y = \frac{\det A_y}{\det A} = \frac{-10}{-15} = \frac{2}{3}}.$$

Exercise 3. $f_1(x, y, z, t) = x^2 - y^2 + zt = -3$, $f_2(x, y, z, t) = e^x + \frac{y}{z} + t^2 = 0$, $P = (0, -2, 1, 1)$.

Solution. (a) $f_1(P) = 0 - 4 + 1 = -3$ (OK), $f_2(P) = 1 + \frac{-2}{1} + 1 = 0$ (OK).

$\boxed{P \text{ satisfies both equations.}}$

(b) IFT on (y, z) (treat x, t as parameters).

$$J_{(y,z)} = \frac{\partial(f_1, f_2)}{\partial(y, z)} = \begin{pmatrix} -2y & t \\ \frac{1}{z} & -\frac{y}{z^2} \end{pmatrix}_P = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det = 7 \neq 0.$$

Hence locally $(y, z) = (y(x, t), z(x, t))$ exists and is C^1 .

$\boxed{\text{Yes (IFT): } J_{(y,z)}(P) \text{ invertible}}$

(c) Use Jacobians and their inverse (no direct differentiation).

$$J_{(x,t)} = \frac{\partial(f_1, f_2)}{\partial(x, t)} = \begin{pmatrix} 2x & z \\ e^x & 2t \end{pmatrix}_P = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

IFT gives

$$\frac{\partial(y, z)}{\partial(x, t)} = -J_{(y,z)}^{-1} J_{(x,t)} = -\frac{1}{7} \begin{pmatrix} 2 & -1 \\ -1 & 4 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = -\frac{1}{7} \begin{pmatrix} -1 & 0 \\ 4 & 7 \end{pmatrix} = \begin{pmatrix} \frac{1}{7} & 0 \\ -\frac{4}{7} & -1 \end{pmatrix}.$$

Thus

$$\boxed{\left(\frac{\partial y}{\partial t}\right)_x(P) = 0}, \quad \left(\frac{\partial y}{\partial x}\right)_t(P) = \frac{1}{7}, \quad \left(\frac{\partial z}{\partial x}\right)_t(P) = -\frac{4}{7}, \quad \left(\frac{\partial z}{\partial t}\right)_x(P) = -1.$$